

Jainta Paul

Salt Lake City, UT — jaintapaul1998@gmail.com — u1471999@utah.edu — pauljainta.github.io/jp

Summary

Former Software Engineer - Pursuing Ph.D. in Computer Science. Built static-analysis platforms, automated security pipelines, and program-analysis tools, and led end-to-end development of systems involving OpenPLC/LLVM instrumentation, machine-learning evaluation pipelines, and agent-based cyber-physical experimentation. Proficient in Python, Java, C++, and JavaScript, with strong experience in software development, automation, security, and systems engineering.

Education

University of Utah, Kahlert School of Computing

Ph.D. Student in Computer Science, Ongoing

Jan 2024 – Present

Salt Lake City, UT

University of Utah, Kahlert School of Computing

M.S. in Computer Science, GPA: 4.00/4.00

May 2026

Salt Lake City, UT

Bangladesh University of Engineering and Technology

B.Sc. in Computer Science and Engineering, GPA: 3.32/4.00

Feb 2017 – May 2022

Dhaka, Bangladesh

Technical Skills

Languages: Python, Java, C++, JavaScript, SQL

Systems & Tools: Git, Docker, Linux, CUDA, PyTorch

Software/Security: Static Analysis, Taint Analysis, Vulnerability Detection, Threat Modeling

Software Engineering Experience

Software Engineer

Jun 2022 – Nov 2023

[OpenRefactory, Inc.](#)

- **Test Drive – OpenRefactory:** Designed and led a GitHub-integrated web platform that allowed users to run static-analysis trials on real-world repositories, with OAuth authentication, repository selection, automated job orchestration, and and result reporting.
- **Open-Source Bug Discovery Pipeline:** Built automated workflows to analyze open-source repositories, identify security and compliance issues, triage findings, and generate reports for maintainers, reducing the amount of manual review needed in the vulnerability reporting process.
- **Intelligent Code Repair:** Improved the Java Intelligent Code Repair engine by implementing call-graph construction, points-to analysis, and static taint tracking, expanding its ability to detect vulnerabilities and generate more precise repairs across large codebases.

Selected Research Projects

ICSTRACKER: Intrusion Backtracking for Industrial Control Systems

Published in DSN 2025

GitHub: github.com/icstracker/icstracker

- Led the MiniWater implementation effort, configuring the OpenPLC-based water-treatment testbed, compiling instrumented controller/runtime code, and validating the full data-collection pipeline across attack scenarios.
- Developed C/C++ and Python instrumentation workflows around OpenPLC and LLVM to capture application-level function traces, runtime values, timestamps, and execution events from PLC control logic.
- Integrated PLC programs, Python PSM/trace-generation scripts, and simulator components to produce structured logs for reconstructing cyber-physical attack paths from physical effects back to controller behavior.

Quantifying Privacy in Edge and Multimodal Systems through Games

Accepted at PETS 2026

GitHub: github.com/iotrustlab/NNEdge-Privacy

- Led development of a experimental framework for evaluating privacy leakage in edge/on-sensor ML systems under different adversarial access models.
- Built preprocessing, feature-extraction, training, and evaluation pipelines for IMU-based activity datasets, supporting binary-output, single-modality, and multi-modality privacy games.
- Implemented adversary models and leakage metrics across decision trees, random forests, CNNs, RNN/LSTMs, and Transformers to measure how interface design choices affect inference risk.

ICS-AGENT: Agent-Based Wargaming for Industrial Control Systems Cyber Security

2025 – Present

- Led development of a cyber-physical wargaming framework that models attacker–defender interaction over an OpenPLC-based water-treatment testbed.
- Implemented telemetry collection, process-state tracking, attacker actions, defender mitigations, and safety-bound evaluation for closed-loop ICS experiments.
- Built scripted, rule-based, and LLM-driven agent workflows and evaluated them across benchmark attack scenarios using detection, mitigation, containment, and safety-violation metrics.

Service & Awards

- Research Mentor, University of Utah REU, Summer 2024.
- Artifact Evaluator, USENIX VehicleSec 2026 and ACSAC 2025.
- Graduate Tuition & Research Scholarship, University of Utah; University Admission Scholarship, BUET.